DEMONSTRATION ONLY — NOT A QUALIFIED PROCEDURE; THE FACILITY OWNER'S QUALIFIED PROCEDURES AND APPLICABLE CODES GOVERN.

WO-500-01 — Replace 3 supports on 8"-P150-FL-2009

Area	A500 Pipe rack	**Task**	SP Pipe support replacement
Priority	3 (reliability)	**Issued by**	Facility owner (demonstration)
Primary asset	rack-bent:720 — Pipe rack bent at E 60'-0" — E 60'-0" N 0.000" EL 122'-0"		
Equipment	—	**Lines**	8"-P150-FL-2009
Isometrics	ISO-FL-2009 (owner's drawings, attached)		
Permits	General work permit, Hot work permit		

Reason

Corroded shoes and guides on the flare header.

Scope of work

1. Support 8"-P150-FL-2009 temporarily either side of each support location.
2. Cut out guide 8"-P150-FL-2009/S1 at E 40'-0" N 4'-6" EL 122'-0"; prepare surfaces.
3. Cut out shoe 8"-P150-FL-2009/S2 at E 60'-0" N 4'-6" EL 122'-0"; prepare surfaces.
4. Cut out guide 8"-P150-FL-2009/S3 at E 80'-0" N 4'-6" EL 122'-0"; prepare surfaces.
5. Weld new supports to WS-03; VT and MT of every fillet weld (SW-NDE-3).
6. Remove temporary supports; check the line rests on every support.

Affected items: 8"-P150-FL-2009/S1, 8"-P150-FL-2009/S2, 8"-P150-FL-2009/S3

Rules: SW-ITP-1, SW-ITP-2, SW-ITP-3, SW-NDE-3, SW-PS-12, SW-WO-1, SW-WO-2, SW-WO-7, SW-WPS-1

Owner's isometric drawings (SW-IMP-6)

Drawing	Line drawn	Work line	Sheets	Match to the plant model
ISO-FL-2009	8"-P150-FL-2009	8"-P150-FL-2009	5	0 error(s), 1 warning(s) — see below

ISO-FL-2009 (PDF): ../isometrics-imported/ISO-FL-2009.pdf — the sheets that hold the work follow the ITP.

Drawing	Sheet	Level	Finding
ISO-FL-2009	1	warning	weld 8"-P150-FL-2009/W2 (E 21'-11 5/16" N 4'-6" EL 123'-0", sheet 1) has no label

Assets in the plant model (SW-WO-7)

Tag	Kind	Description	Line	Location
rack-bent:720	structure	Pipe rack bent at E 60'-0"		E 60'-0" N 0.000" EL 122'-0"
8"-P150-FL-2009/S1	support	GUIDE FOR 8" LINE (GUIDE-8)	8"-P150-FL-2009	E 40'-0" N 4'-6" EL 122'-0"
8"-P150-FL-2009/S2	support	SHOE FOR 8" LINE (SHOE-8)	8"-P150-FL-2009	E 60'-0" N 4'-6" EL 122'-0"
8"-P150-FL-2009/S3	support	GUIDE FOR 8" LINE (GUIDE-8)	8"-P150-FL-2009	E 80'-0" N 4'-6" EL 122'-0"

Parts

Description	Qty	Replaces
GUIDE FOR 8" LINE (GUIDE-8)	1	8"-P150-FL-2009/S1
SHOE FOR 8" LINE (SHOE-8)	1	8"-P150-FL-2009/S2
GUIDE FOR 8" LINE (GUIDE-8)	1	8"-P150-FL-2009/S3

New welds (SW-WO-4)

Weld	Replaces	On drawing	Line	Size	Wall	Shop/field	Spec	NDE	Location	Welder	Date
WO-500-01-W01	8"-P150-FL-2009/S1	—	8"-P150-FL-2009	8" P150	—	field	WS-03	VT + MT	E 40'-0" N 4'-6" EL 122'-0"		
WO-500-01-W02	8"-P150-FL-2009/S2	—	8"-P150-FL-2009	8" P150	—	field	WS-03	VT + MT	E 60'-0" N 4'-6" EL 122'-0"		
WO-500-01-W03	8"-P150-FL-2009/S3	—	8"-P150-FL-2009	8" P150	—	field	WS-03	VT + MT	E 80'-0" N 4'-6" EL 122'-0"		

Welding specification WS-03 — Fillet welds of support attachments

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Scope	Shoes, guides and anchors to pipe or steel
Joint	Fillet, leg per the support part (SW-PS-12)
Process	Shielded metal arc throughout
Filler	SW-F2 covered electrode
Preheat	≥ 30 °F above dew point and ≥ 60 °F
Interpass	≤ 450 °F
Post-weld heat treatment	None
Rules	SW-WPS-1, SW-WPS-2, SW-WPS-3, SW-PS-12

NDE

Method	Extent	Welds	Basis
VT + MT	100 % of attachment fillet welds	WO-500-01-W01, WO-500-01-W02, WO-500-01-W03	SW-NDE-3

Inspection and test plan (SW-ITP-1..3)

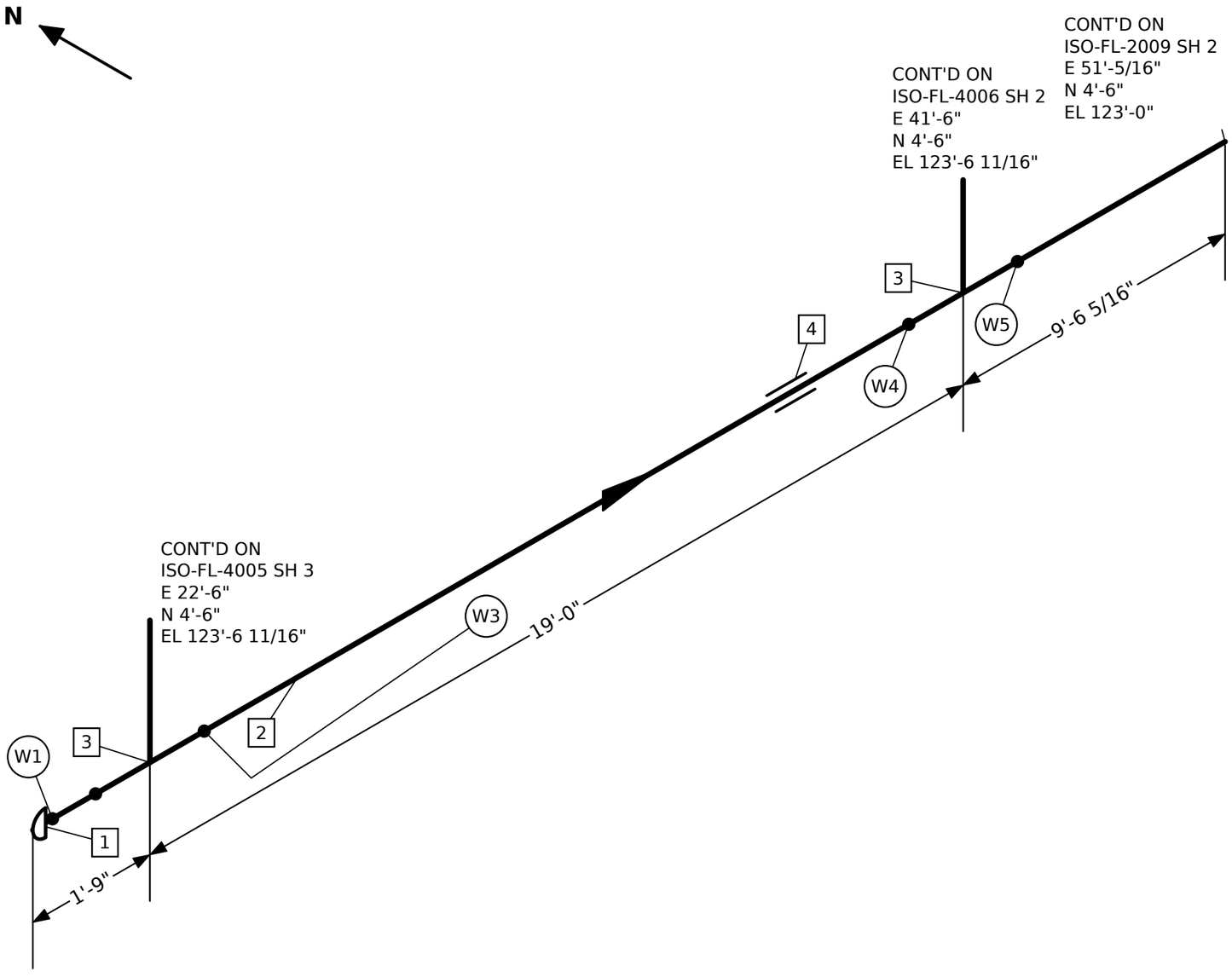
H hold · W witness · R review · S surveillance. Status: accepted / rejected / not applicable.

#	Activity	Reference	Acceptance	Record	Contr.	Owner	3rd party	Status	Signed (contr. / owner / 3rd)	Date	Remark
1	Temporary supports in place; line load taken	Owner rigging procedure	Line held at its elevation	Daily report	H	W	—				
2	Old supports removed; pipe surface inspected under the support	SW-NDE-4	No wall loss beyond the owner's criteria (else raise a spool work order)	Inspection record	W	H	—				
3	New supports welded to WS-03	WS-03, SW-WPS-3	Fit and leg as specified	Weld record	S	S	—				
4	VT and MT of every attachment weld	SW-NDE-3	No crack or lack of fusion	VT/MT reports	W	R	R				
5	Records complete; work order handed back to operations	SW-ITP-3	All ITP activities accepted or not applicable	Signed ITP and completion record	H	H	R				

Where the work is on the owner's drawings (SW-IMP-8)

Each affected item, replaced weld and isolation point, with the drawing and sheet that hold it and the weld label printed there. The sheets follow, copied unchanged.

Tag	Role	Drawing	Sheet	Weld label
8"-P150-FL-2009/S1	affected	ISO-FL-2009	1	
8"-P150-FL-2009/S2	affected	ISO-FL-2009	2	
8"-P150-FL-2009/S3	affected	ISO-FL-2009	2	



BILL OF MATERIALS

PT	QTY	NPS	DESCRIPTION
1	1	8"	CAP 8" P150
2	27'-9 5/8"	8"	PIPE 8" P150 BEVELLED ENDS
3	2	8"x4"	TEE REDUCING 8" × 4" P150
4	1	8"	GUIDE FOR 8" LINE (GUIDE-8)

REV	DATE	DESCRIPTION	BY
0	2026-11-01	ISSUED FOR CONSTRUCTION	-

COMPANY

SMARTWELD

TITLE

FLARE AND RELIEF OUTLETS

NOTES:

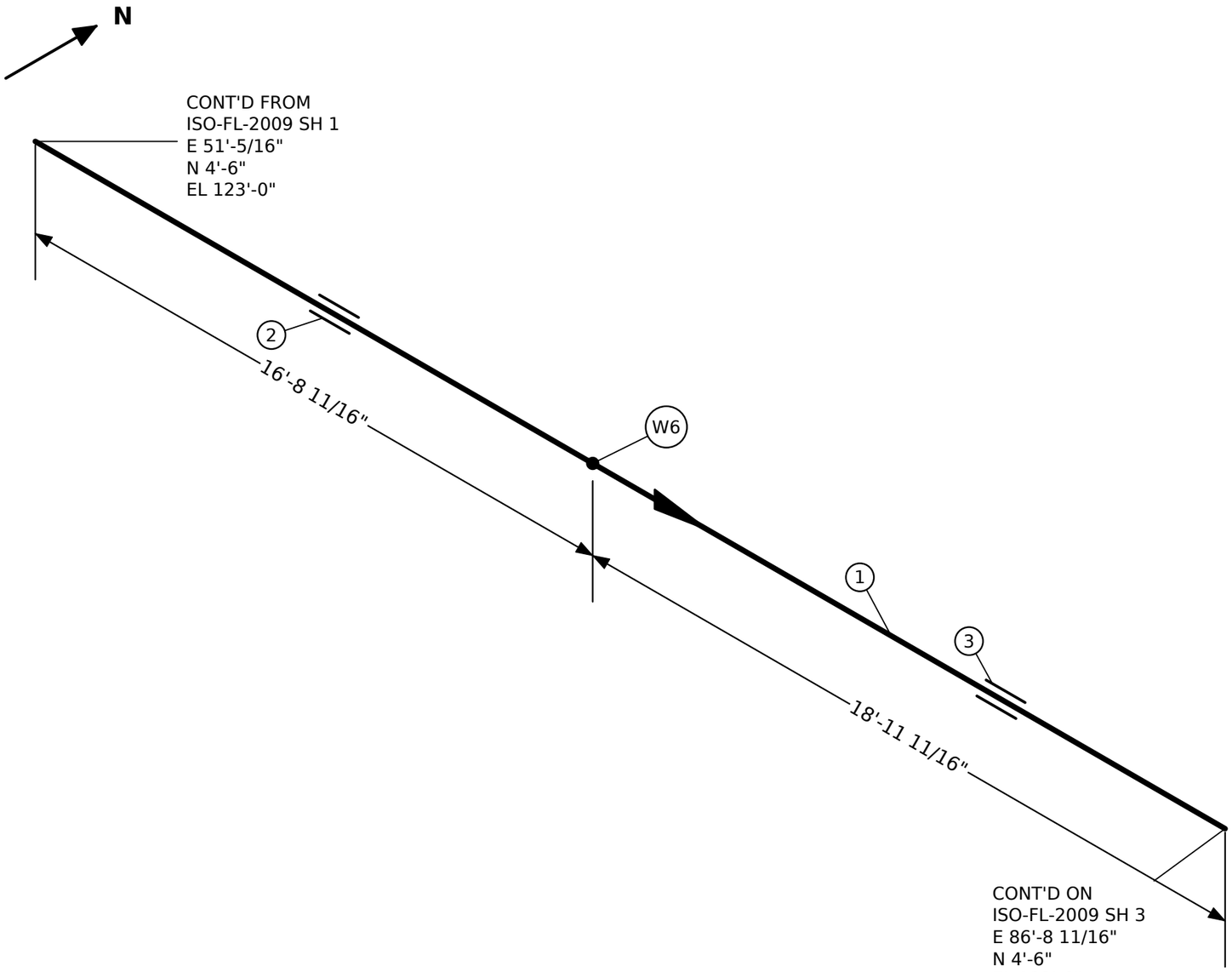
1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD

TEST PRESS.	INSULATION	PAINT
-	NONE	-
PWHT	NDE	FLUID
-	-	FL

LINE NO.	PIPING CLASS	DESIGN PRESS.	DESIGN TEMP.
8"-P150-FL-2009	F150	150 PSIG	-

DRAWING NO.	SHEET	REV
ISO-FL-2009	1 OF 5	0

DATE	DRAWN	CHECKED	SCALE
2026-11-01	-	-	NTS



BILL OF MATERIALS

PT	QTY	NPS	DESCRIPTION
1	35'-8 3/8"	8"	PIPE 8" P150 BEVELLED ENDS
2	1	8"	SHOE FOR 8" LINE (SHOE-8)
3	1	8"	GUIDE FOR 8" LINE (GUIDE-8)

REV	DATE	DESCRIPTION	BY
0	2026-11-01	ISSUED FOR CONSTRUCTION	-

COMPANY

SMARTWELD

TITLE

FLARE AND RELIEF OUTLETS

NOTES:

1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD

TEST PRESS.	INSULATION	PAINT
-	NONE	-
PWHT	NDE	FLUID
-	-	FL

LINE NO.	PIPING CLASS	DESIGN PRESS.	DESIGN TEMP.
8"-P150-FL-2009	F150	150 PSIG	-

DRAWING NO.	SHEET	REV
ISO-FL-2009	2 OF 5	0

DATE	DRAWN	CHECKED	SCALE
2026-11-01	-	-	NTS

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